

About Me

I am a Ph.D. candidate at Georgia Tech, advised by Profs. Vijay Ganesh and Sriram Vishwanath. I work at the intersection of **automated reasoning** and machine learning, building **neuro-symbolic AI** that couples large language models (LLMs) with formal methods, solvers, and verifiers. My expertise spans **LLM post-training** via **reinforcement learning from symbolic feedback**, **semantic alignment**, and **evolutionary search**, applied broadly across **AI for software engineering (AI4SE)** and **AI for mathematics (AI4Math)**: **self-improving coding agents** that evolve their own harnesses, and auto-formalization and proof synthesis in Lean. My work has appeared at ICLR, ICSE, IJCAI, ECAI, and WWW, including a Distinguished Paper Award at ICSE 2026.

Previously, I earned my Master's in Computer Science from IIT Kharagpur (Institute Silver Medalist, 2022) and my Bachelor's from Jadavpur University (University Gold Medalist, 2020), graduating first in my class at both; I also spent the first year of my Ph.D. at the University of Waterloo, Canada. I have interned twice at AWS AI Labs (Amazon, Seattle) as an Applied Scientist, working on post-training verifier-guided coding agents, self-improving agents, and multi-agent scientific discovery. I am a recipient of the Reliance Foundation Scholarship for AI & CS (2021) and the national JBNSTS, KVPY, and NTSE awards from the Government of India.

Research Areas: AUTOMATED REASONING · NEURO-SYMBOLIC AI · CODING AGENTS · FORMAL METHODS · AI FOR MATH · LLMs

Technical Skills

- **Languages:** Python, C/C++, Java, Lean 4, MATLAB, JS
- **Post-Training:** RLVR & SFT (TRL), PyTorch, DeepSpeed
- **Verification:** Lean 4/Mathlib, SAT/SMT, symbolic execution
- **Compilers/PL:** LLVM-IR, lex/flex, yacc/bison
- **Agents:** harness design & evals, MCP, Claude Code/Codex

Languages

- **English:** Full Professional Working Proficiency
- **Hindi:** Full Professional Working Proficiency
- **Bengali:** Native or Bilingual Proficiency

Education

Georgia Institute of Technology (Georgia Tech), ATLANTA, USA

Ph.D. in Computer Science

– Advisor: Profs. Vijay Ganesh & Sriram Vishwanath

Aug 2023–Present

GPA: 4.0/4.0

University of Waterloo, ONTARIO, CANADA

Ph.D. in Computer Science (Transferred)

– Advisor: Prof. Vijay Ganesh

2022–2023

Grade: 90.33%

Indian Institute of Technology (IIT) Kharagpur, INDIA

M.Tech in Computer Science & Engineering

- ★ **Institute Silver Medalist**, Department Rank: 1st/75
- Thesis: “Fairness in Virtual Conference Scheduling” [PDF] (*Best Master's Thesis Nomination*)
- Advisor: Prof. Niloy Ganguly

2020–2022

CGPA: 9.84/10

Jadavpur University, KOLKATA, INDIA

B.E. in Computer Science & Engineering

- ★ **University Gold Medalist**, Department Rank: 1st/70
- Thesis: “A Real-Time Semantic Image Segmentation Approach Through Region-Partitioning” [PDF]
- Advisor: Prof. Sanjoy K. Saha

2016–2020

CGPA: 9.57/10

Work Experience

Amazon Web Services (AWS), SEATTLE, USA

Applied Scientist II Intern · Kiro Science Team, AWS AI Labs

- Manager: Purak Jain · Mentor: Nikos Kanakaris
- Leading R&D of **MILO**, an evolutionary search framework for automated harness discovery and self-improving coding agents via multi-agent orchestration; **first-author preprint** [arXiv] [Project Page].
- New SoTA on *Terminal-Bench 2.1*, *PaperBench* and *DeepSWE*; 3 new records on open math problems.

May 2026 – Present

Applied Scientist II Intern · Next Gen DevX (NGDE) Team, AWS AI Labs

- Manager: Bhavana Bhasker · Mentor: Sam Davidson
- Led R&D of **TerraFormer**, a neuro-symbolic coding agent for Infrastructure-as-Code; earned the **highest performance rating** (*Exceeded Expectations*) and a **nomination** for Georgia Tech's *Intern of the Year*.
- Published a **first-author paper** at **ICSE 2026** (CORE A* conf.); won a **Distinguished Paper Award**.

Publications

G : [Google Scholar](#) (As of Sep 2026: citations = 288, h-index = 8, i10-index = 6)

♦: A or A* venues (prestigious and highly-selective) per the CORE Computer Science Conference Rankings

PEER-REVIEWED INTERNATIONAL CONFERENCE PAPERS

- [13] **ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings** ♦ ICLR 2026
P. Jana, K. Kale, A. E. Tanriverdi, C. Song, S. Vishwanath, and V. Ganesh
14th Int'l Conference on Learning Representations, Rio de Janeiro, Brazil, 2026. (Acpt. Rate: 28%) [\[Link\]](#) [\[arXiv\]](#)
Also accepted for presentation (non-archival) at the *9th Int'l Symposium on AI Verification (SAIV)* and the *Workshop on AI for Mathematics & Computer Science Research (AIMACS)*, FLoC 2026, Lisbon, Portugal.
- [12] **TerraFormer: Automated IaC with LLMs Fine-Tuned via Policy-Guided Verifier Feedback** ♦ ICSE 2026
P. Jana, S. Davidson, B. Bhasker, A. Kan, A. Deoras, and L. Callot
IEEE/ACM 48th Int'l Conf. on Software Engineering, SEIP Track, Rio de Janeiro, Brazil, 2026. [\[Link\]](#) [\[arXiv\]](#) [\[YouTube\]](#)
(★ Distinguished Paper Award)
- [11] **RLSF: Fine-tuning LLMs via Symbolic Feedback** ♦ ECAI 2025
P. Jha, P. Jana, P. Suresh, A. Arora, and V. Ganesh
28th European Conf. on Artificial Intelligence, Bologna, Italy, pp. 1687–94, 2025. (Acpt. Rate: 23%) [\[Link\]](#) [\[arXiv\]](#)
- [10] **CoTran: LLM-based Code Translator using RL with F/b from Compiler and Symbolic Exec.** ♦ ECAI 2024
P. Jana, P. Jha, H. Ju, G. Kishore, A. Mahajan, and V. Ganesh
27th European Conf. on Artificial Intelligence, Santiago, Spain, pp. 4011–18, 2024. (Acpt. Rate: 23%) [\[Link\]](#) [\[arXiv\]](#)
- [9] **NeuroSymbolic LLM for Mathematical Reasoning and Software Engineering** ♦ IJCAI 2024
P. Jana
33rd Int'l Joint Conf. on Artificial Intelligence, Doctoral Consortium, pp. 8492–93, 2024. (Acpt. Rate: 17%) [\[Link\]](#)
- [8] **Scheduling Virtual Confs. Fairly: Achieving Equitable Participant and Speaker Satisfaction** ♦ WWW 2022
G. K. Patro, P. Jana, A. Chakraborty, K. P. Gummadi, and N. Ganguly
31st ACM Web Conference, Lyon, France, pp. 2646–2656, 2022. (Acpt. Rate: 17.7%) [\[Link\]](#) [\[PDF\]](#)
- [7] **Unsupervised Action Localization Crop in Video Retargeting for 3D ConvNets** TENCON 2021
P. Jana, S. Bhaumik, and P. P. Mohanta
2021 IEEE Region 10 (Asia-Pacific) Conference, Auckland, New Zealand, pp. 670–675, 2021. [\[Link\]](#) [\[PDF\]](#)
- [6] **AutoDrone: Shortest Optimized Obstacle-Free Path Planning for Autonomous Drones** ICDC 2021
P. Jana and D. Jana
International Conf. on Computational Intelligence, Data Science & Cloud Computing, pp. 1–13, 2021. [\[Link\]](#) [\[PDF\]](#)
- [5] **Two Stage Semantic Segmentation by SEEDS & Fork Net** CALCON 2020
A. Mukherjee, P. Jana, S. Chakraborty, and S. K. Saha
2020 IEEE Calcutta Conference, pp. 283–287, 2020. [\[Link\]](#) [\[PDF\]](#)
(★ Best Student Paper 2nd Runner-Up Award)
- [4] **A Multi-tier Fusion Strategy for Event Classification in Unconstrained Videos** PReMI 2019
P. Jana, S. Bhaumik, and P. P. Mohanta
8th International Conf. on Pattern Recognition & Machine Intelligence, Springer, pp. 515–524, 2019. [\[Link\]](#) [\[PDF\]](#)

- [3] **Key-Frame based Event Recognition in Unconstrained Videos using Temporal Features** TENSYP 2019
P. Jana, S. Bhaumik, and P. P. Mohanta
2019 IEEE Region 10 (Asia-Pacific) Symposium, pp. 349–354, 2019. [Link] [PDF]
- [2] **Handwritten Document Image Binarization: An Adaptive K-means based Approach** CALCON 2017
P. Jana, S. Ghosh, S. K. Bera, and R. Sarkar
2017 IEEE Calcutta Conference, pp. 226–230, 2017. [Link] [PDF]
- [1] **A Fuzzy C-Means Based Approach Towards Efficient Document Image Binarization** ICAPR 2017
P. Jana, S. Ghosh, R. Sarkar, and M. Nasipuri
9th International Conference on Advances in Pattern Recognition (ICAPR), pp. 332–337, 2017. [Link] [PDF]

INVITED PAPERS

- [1] **Semantic Alignment Models: A Bridge Between Informal and Formal Mathematics** CICM 2026
P. Jana, S. Vishwanath, and V. Ganesh
19th Conference on Intelligent Computer Mathematics, Invited Paper, Ljubljana, Slovenia, 2026.

PEER-REVIEWED WORKSHOP PAPERS

- [2] **Learning Flexible Context Management Unlocks Test-Time Scaling** NeurIPS · LCFM 2026
X. Li, M. Goswami, H. Liu, N. Kanakaris, L. Huang, P. Jana, P. Blöbaum, and P. Jain
Workshop on Long Context Foundation Models (LCFM) @ NeurIPS, Atlanta, USA, 2026. [Site]
- [1] **On the Off-Policy Teacher in On-Policy Distillation** NeurIPS · MATH-AI & Pre-to-Post 2026
L. Huang, H. Liu, M. Goswami, X. Li, P. Jana, N. Kanakaris, P. Blöbaum, and P. Jain
6th Workshop on Mathematical Reasoning and AI (MATH-AI) @ NeurIPS, Atlanta, USA, 2026. [Site]
& Workshop on Transitioning from Pre-Training to Post-Training (Pre-to-Post) @ NeurIPS, Sydney, Australia, 2026. [Site]

JOURNAL PAPERS / BOOK CHAPTERS

- [3] **SALiEnSeA: Spatial Action Localization & Temporal Attention for Video Event Recognition** IJCISIM 2022
P. Jana, S. Bhaumik, and P. P. Mohanta
Int'l Journal of Computer Info. Systems and Industrial Mgmt. Appl., vol. 14, pp. 270–284, MIR Labs, 2022. [Link] [PDF]
- [2] **Recent Trends in 2D Object Detection and Applications in Video Event Recognition** River 2022
P. Jana and P. P. Mohanta
Advances of Deep Learning and its Appl. in Object Detection and Recognition, River Publishers, 2022. [Link] [PDF]
- [1] **Event and Activity Recognition in Video Surveillance for Cyber-Physical Systems** Springer 2021
S. Bhaumik, P. Jana, and P. P. Mohanta
Emergence of Cyber Physical System & IoT in Smart Automation and Robotics, Springer Nature, 2021. [Link] [PDF]

PREPRINTS / UNDER SUBMISSION

- [2] **Abstractions-of-Thought: Intermediate Repr. for LLM Reasoning in Hardware Design** arXiv 2025
M. DeLorenzo, K. Tieu, P. Jana, P. Jha, D. Kalathil, V. Ganesh, and J. Rajendran
arXiv preprint arXiv:2505.15873, 2025. [arXiv]
- [1] **Reducing Opinion Echo-Chambers by Intelligent Placement of Moderate-Minded Agents** arXiv 2023
P. Jana, R. R. Choudhury, and N. Ganguly
arXiv preprint arXiv:2304.10745, 2023. [arXiv]

Scholarships and Awards

DURING PH.D.

- **Distinguished Paper Award, ICSE-2026** 2026
One of 3 papers recognized among 74 accepted in the SEIP track (IEEE/ACM 48th Int'l Conf. on Software Engineering)

- **2nd Place, College of Computing (CoC) Graduate Poster Symposium, Georgia Tech** 2025
Runner-up at the college-wide graduate research symposium across SCS, SCP, IC, CSE; evaluated by faculty
- **Top Reviewer, NeurIPS-2024** 2024
Top 8% of reviewers (1,300 out of 15,160); awarded complimentary full conference registration
- **Lunch with a EurAI Fellow Invitee, ECAI-2024** 2024
Invited by ECAI-2024 for a select Ph.D. lunch discussion with Prof. Mehdi Dastani (EurAI Fellow; Chair, ECAI-2025)

DURING MASTER'S

- **Institute Silver Medalist, CSE Department Topper** 2022
Highest GPA in M.Tech. Computer Science, IIT Kharagpur; Rank 1st/75
- **Best M.Tech. Thesis Award Nominee, CSE Department** 2022
Nominated for Best Thesis, M.Tech. Computer Science, IIT Kharagpur
- **Reliance Foundation Scholarship for AI & Computer Science** 2021
One of 38 Master's students selected across India; INR 0.6 million in academic funding
- **CSI Young IT Professional Award – East Region Runner-Up** 2021
Runner-up in the Eastern India Region; youngest participating team in the competition
- **GATE Scholar (Graduate Aptitude Test in Engineering)** 2020
All-India Rank 188/100,000+ (99.81 percentile); stipend of INR 12,400/month for 2 years

DURING BACHELOR'S & HIGH-SCHOOL

- **University Gold Medalist, CSE Department Topper** 2020
Highest GPA in B.E. Computer Science, Jadavpur University; Rank 1st/70
- **Top 6 in TIFR GS-2020 (Computer Science) Nationwide** 2020
Among 6 students selected across India for direct Ph.D. admission at TIFR, India via competitive exam and interview
- **CSE Department Annual Gold Medalist** 2017–2019
Highest annual GPA in CSE, Jadavpur University, for three consecutive years
- **Summer Research Fellowship Programme (SRFP)** 2018, 2019
Selected by the Indian Science Academies among the top 350 undergrads nationwide; two consecutive years
- **Special Mention Award, Computer Vision Summer School (SSCVGIP)** 2018
Awarded by the Indian Statistical Institute and IUPRAI for best project at the Computer Vision Summer School
- **JBNSTS Senior Scholar (Jagadis Bose National Science Talent Search)** 2016
One of 47 selected across Science, Engineering, and Medicine; among India's oldest, most selective science talent awards
- **KVPY Awardee (Kishore Vaigyanik Protsahan Yojana)** 2015
All-India Rank 143/100,000+ (top 0.14%); prestigious national fellowship awarded by Govt. of India and IISc Bangalore
- **NTSE Scholar (National Talent Search Examination)** 2012
Top 1,000 out of ~1,000,000 examinees (top 0.1%); Govt. of India national scholarship for academic excellence

Invited Talks

- **Invited Talk at School of Computing Science, Simon Fraser University, CANADA** Nov 2025
ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings
- **Invited Talk at Mathematical AI Seminar, Fields Institute, University of Toronto, CANADA** Oct 2025
AI for Math: Neuro-Symbolic Auto-Formalization into Lean via Joint Embeddings [[Abstract](#)] [[YouTube](#)]
- **Technical Track Talk at 25th High Confidence Software & Systems Conference (HCSS), USA** May 2025
Neuro-Symbolic Techniques for LLM-based Code Generation and Auto-Formalization of Proofs [[Slides](#)] [[Proceedings](#)]
- **Invited Talk at ACM Student Chapter, Heritage Institute of Technology, Kolkata, INDIA** Jan 2025
Neuro-Symbolic Approaches for Fine-Tuning Code Language Models [[Flyer](#)]
- **Invited Talk at Jagadis Bose Scholars Professional Development Forum** May 2021
Counterfactual Fairness in Machine Learning [[YouTube](#)] [[Flyer](#)]

Teaching Experience

- **Co-Organizer & Speaker, PLSE/FM + AI Workshop** [[Site](#)], Georgia Tech · Google · Oak Ridge National Lab · GTRI Apr 2026
Delivered a tutorial on Neuro-Symbolic AI for Code & Math to 150+ PhD students, faculty, and industry professionals.

Co-Organizer & Instructor , IDEaS One-Day Tutorial on AI4Science [Tutorial Site], Georgia Tech, USA Delivered lectures and hands-on sessions on LLMs and formal methods to 80+ Ph.D. students, postdocs, and scientists.	Oct 2025
Teaching Assistant , Georgia Tech, USA CS 7001: <i>Introduction to Graduate Studies</i> (Profs. Betsy DiSalvo & Kuldeep S. Meel; 60+ students) CS 3600: <i>Introduction to Artificial Intelligence</i> (Mr. Sourish Dhekane; 120+ students)	Fall 2026 · Summer 2024
Teaching Assistant , University of Waterloo, CANADA CS 135: <i>Designing Functional Programs</i> (Dr. Byron Weber Becker; 1,200+ students) CS 105: <i>Introduction to Computer Programming</i> (Dr. Muddassir Malik; 60+ students) CS 240: <i>Data Structures and Data Management</i> (Prof. Éric Schost; 400+ students)	Summer 2023 · Spring 2023 · Fall 2022
Teaching Assistant , IIT Kharagpur, INDIA CS 19001: <i>Programming & Data Structures Lab</i> (Drs. Abir Das & Saptarshi Ghosh; 100+ students) CS 60075: <i>Natural Language Processing</i> (Prof. Sudeshna Sarkar; 120+ students)	Fall 2021

Academic Services & Professional Activities

IEEE Student Member Member ID: 92577001	2017 – present
Lead Student Organizer, Foundations of AI (FoAI) Seminar Series [Seminar Page], Georgia Tech, USA Lead organizer for bi-weekly talks hosted by Profs. V. Ganesh, D. Sherrill, and C. Lin; single-handedly manage all logistics: coordinating with visiting faculty and industry researchers, venue, catering, and announcements.	2025 – present
Conference & Journal Service, Participation & Travel Grants PC Member: AAAI (2027, 2026, 2025, 2024, 2023 [AIES]) · ICLR (2027, 2026, 2025, 2024) · / Reviewer ICML (2026, 2025) · NeurIPS (Main + D&B) (2026, 2025, 2024 [Top Reviewer], 2023, 2021) · AISTATS (2027, 2026, 2025) · LICS (2026) · ECAI (2025) · CAV (2025) · NeSy (2024, 2023) · PLDI (2024) · SAT (2024) · CP (2024) · KDD (2022) · ACM TOPLAS (2026) · TMLR (2026, 2025) · Elsevier ASC (2025) · Nature SR (2024) · IEEE SMC (2023) · IEEE Access (2022) · Springer JCST (2020) Attended: ICLR 2026 (Brazil) · ICSE 2026 (Brazil) · HCSS 2025 (USA) · NeurIPS 2024 (Canada) · ECAI 2024 (Spain) · IJCAI 2024 (South Korea) · NeurIPS 2023 (USA) Travel Grants: ICLR 2026 (Brazil) · IJCAI 2024 (South Korea) · CAV 2024 (Canada)	2020 – present

Academic Research Internship Experience

Indian Institute of Technology (IIT) Kharagpur , INDIA <i>Research Intern · Video Action Recognition & Zero-Shot Temporal Detection</i> - Advisors: Dr. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab) - Extended R-C3D to achieve zero-shot end-to-end temporal activity detection in untrimmed videos; tackled recognition and localization of human activities under open-vocabulary label settings.	Apr 2021 – Aug 2022
University of Alberta , CANADA <i>Research Intern · 3D Object Detection & Pose Estimation in Indoor Scenes</i> - Advisor: Prof. Nilanjan Ray (University of Alberta) - Detected and estimated poses of 3D objects in indoor scenes from RGB-D images and point clouds using PointNet and Transformer architectures.	Jun – Dec, 2021
Indian Statistical Institute & IUPRAI , INDIA <i>Research Project Trainee & Summer Intern · Spatio-Temporal Event/Activity Recognition in Videos</i> - Advisor: Dr. Partha Pratim Mohanta (ISI Kolkata) - Awarded: ‘Special-Mention’ at the 5th Summer School on CV, Graphics & Image Processing (2018) - Proposed a hierarchical decision-fusion framework combining frame- and video-level CNN & LSTM features to capture short- and long-term motion; designed a graph-based key-frame selection algorithm.	May – Jul, 2018 & 2019

Summer Research Intern · Carry-less Karatsuba Multiplication over Galois Field in Block Ciphers

- Advisor: Dr. Debrup Chakraborty (ISI Kolkata)
- Supported & funded by: Microsoft Research India · Selected among: 25 students nationwide
- Implemented carry-less Karatsuba multiplication of degree-127 binary polynomials via 64-bit operations and fast polynomial reduction, accelerating AES-GCM in high-performance symmetric-key block ciphers.

Jadavpur University, INDIA

May 2017 – Oct 2018

Undergraduate Researcher · Document Analysis & Medical Image Segmentation

- Advisors: Dr. Ram Sarkar⁽¹⁾, Profs. Mita Nasipuri⁽²⁾ & Subhadip Basu⁽³⁾ (Jadavpur University)
- Proposed background inpainting and fuzzy clustering for binarizing degraded documents^(1,2); developed binarization of 3D microscopic mouse brain images^(1,3) for dendritic spine analysis.

Relevant Coursework Pursued

[During Ph.D.] Georgia Institute of Technology, USA & University of Waterloo, CANADA

2022 – 2025

(CS 6241) Advanced Compiler Optimization · Prof. Santosh Pande
(CS 8903-SAT) SAT and SMT Solvers: Theory, Practice, and Applications · Prof. Vijay Ganesh
(CS 6340) Software Analysis and Test · Prof. Qirun Zhang
(CS 6601) Artificial Intelligence · Prof. Thomas Ploetz
(CS 8903-LCS) Logic in Computer Science · Prof. Suguman Bansal
(CS 846) Advanced Topics in Software Engineering: Requirements Engineering · Prof. Daniel Berry
(CS 898) Advanced Special Topics in Computer Science: Computer Vision · Prof. Yuri Boykov
(CS 685) Machine Learning: Statistical & Computational Foundations · Prof. Shai Ben-David

[During Master's] IIT Kharagpur, INDIA (Total Credit Hours: 49)

2020 – 2022

Algorithm Design & Analysis, AI, NLP, Foundations of CS, Info Retrieval, Complex Networks, High-Performance Computer Architecture, Computational Geometry, Deep Learning, Distributed Systems

[During Bachelor's] Jadavpur University, INDIA (Total Credit Hours: 193)

2016 – 2020

Digital Logic & Circuits, Data Structures & Algo, Design & Analysis of Algo, Numerical Methods, OOP, Microprocessors & Assembly Lang, Graph Theory & Combinatorics, Computer Organization & Architecture, Systems Programming, OS, Formal Lang & Automata, Digital Communication, Computer Graphics, DBMS, Principles of Programming Lang, Compiler Design, Computer Networks, Software Engg., AI, Internet Tech, Data Mining, VLSI, Optimization Tech & Operation Research, NLP, Soft Computing